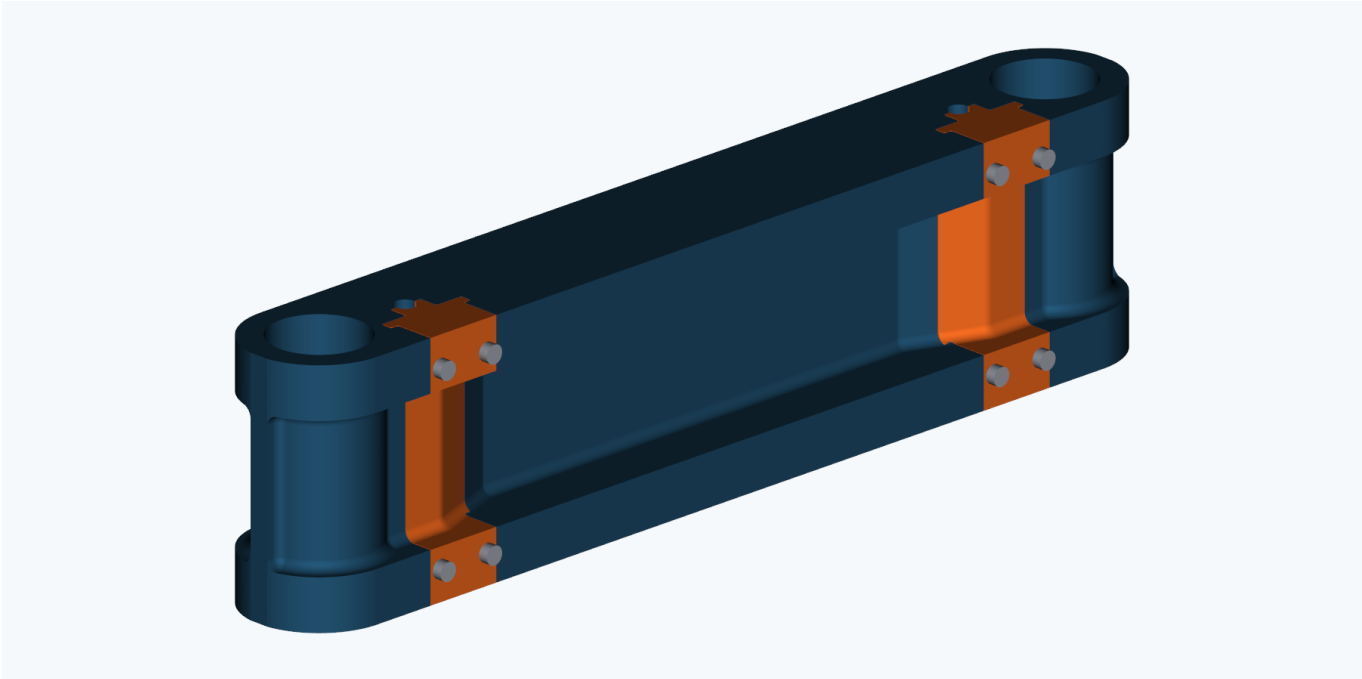


Selected-run evidence status: completed_unassessed

Geometry



Recorded assembly geometry. Original geometry/assembly overview from this selected case's matched report.

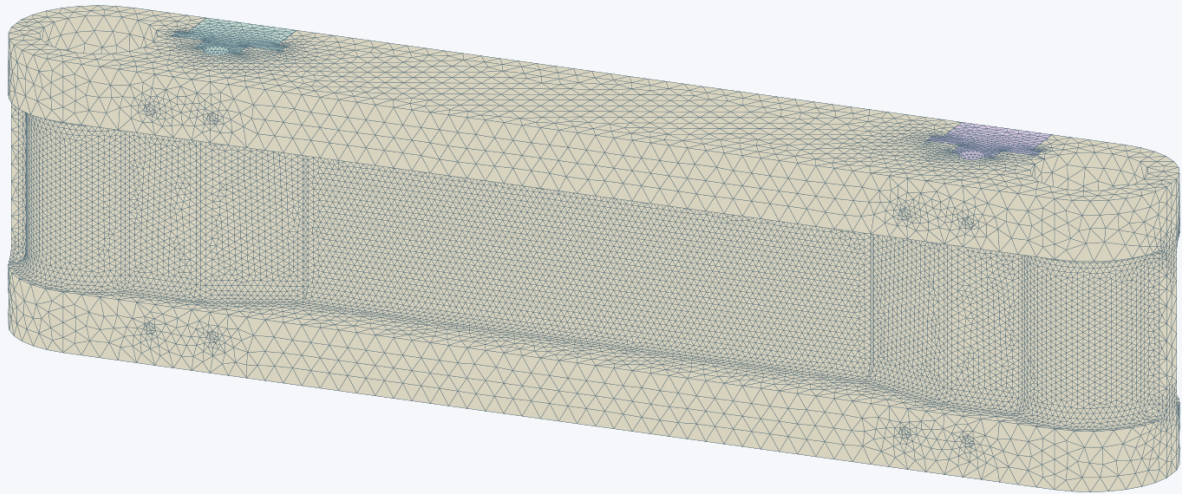
Mesh

Mesh overview and local detail

Mesh nodes	76873	
Tetrahedral elements	321831	
Meshed parts	3	
Component target sizes	5 / 3	mm
Minimum mean-ratio quality	0.0742	0-1

Mesh overview

Undeformed exterior mesh

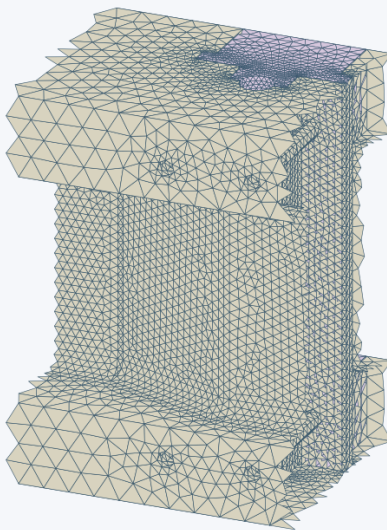


Exterior mesh faces • undeformed geometry

Mesh overview. Undeformed exterior topology of the selected case, verified against its recorded solved-mesh hashes.

Local mesh detail

Local mesh detail



Exterior mesh faces • undeformed geometry

Local mesh detail. Local view of the selected mesh; no local refinement claim is made.

Mean-ratio quality describes element shape and does not establish stress or time-step convergence.

Setup

Recorded selected-case model

Initial impact velocity	50	m/s
Requested simulation duration	8	ms
Foam depth	203.2	mm
Foam width	41.275	mm
Foam height	152.4	mm
Foam element size	10	mm
Initial contact gap	0.1	mm
Contact friction coefficient	0.1	dimensionless
Foam curve scale	1	dimensionless
Time-step scale	0.6	dimensionless
Requested state-output interval	0.05	ms
Startup mass-scaling DT2MS	-1.5e-07	s
Termination time-step ratio	0.01	dimensionless
Printed parts mass	0.5627708	kg
Hardware mass	0.03400217	kg
Motor mass	0.16	kg
Moving mass	0.756773	kg
Initial kinetic energy	945.9662	J

Recorded material, contact and boundary assumptions

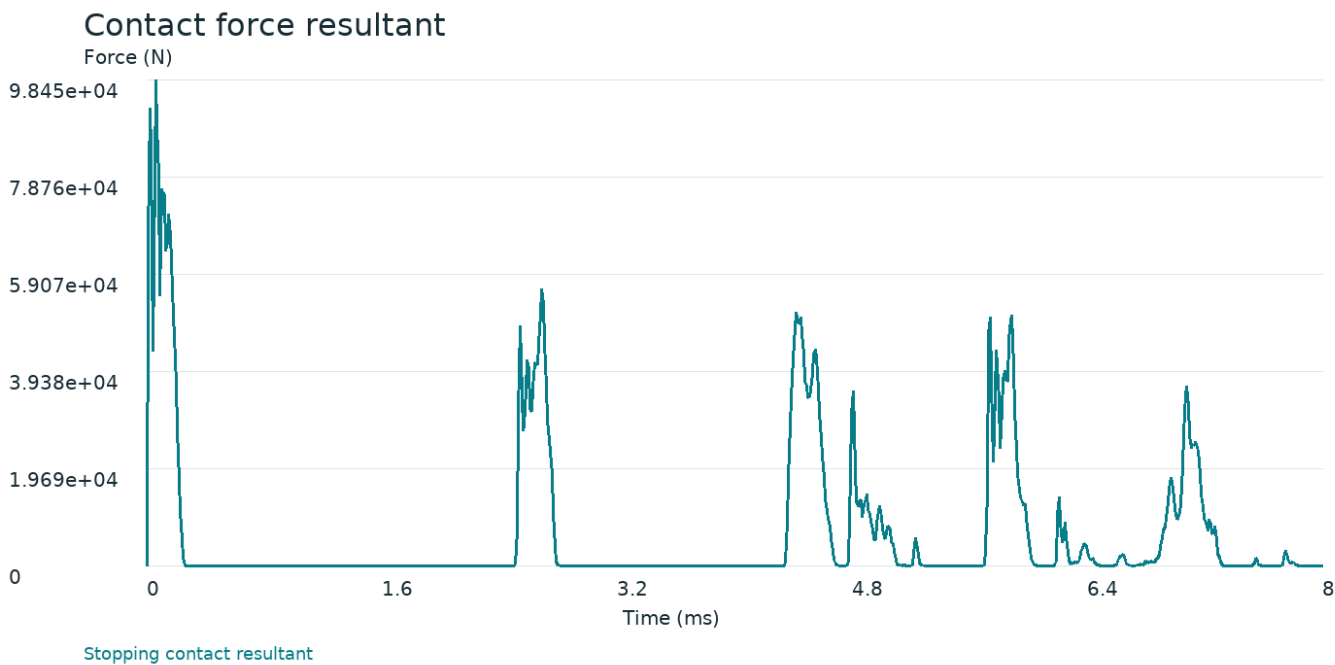
- Unchanged prior clamshell fixture and synthetic 54.46 kg/m³ MAT057 foam; not the proposed 30 kg/m³ product.
- Three independent printed solids; body-to-closure frictional contact plus eight ideal bolt collars; no slip/preload/failure capacity.
- Eight CAD generic steel bolt/nut envelope masses distributed on the collar nodes.
- Two 80 g motor masses distributed on the original motor bores; no casing stiffness.
- PA12-CF10 isotropic linear elastic screen; not printed anisotropic strength qualification.
- Ideal transverse cable-bore guidance leaves axial Z free; no cable compliance, tension or clearance contact.
- Two stationary aluminum spreaders, 0.1 mm initial gap; fixed foam rear Z, free foam sides.

- No carrier, motor thrust, element failure, fracture or damage.

Solve

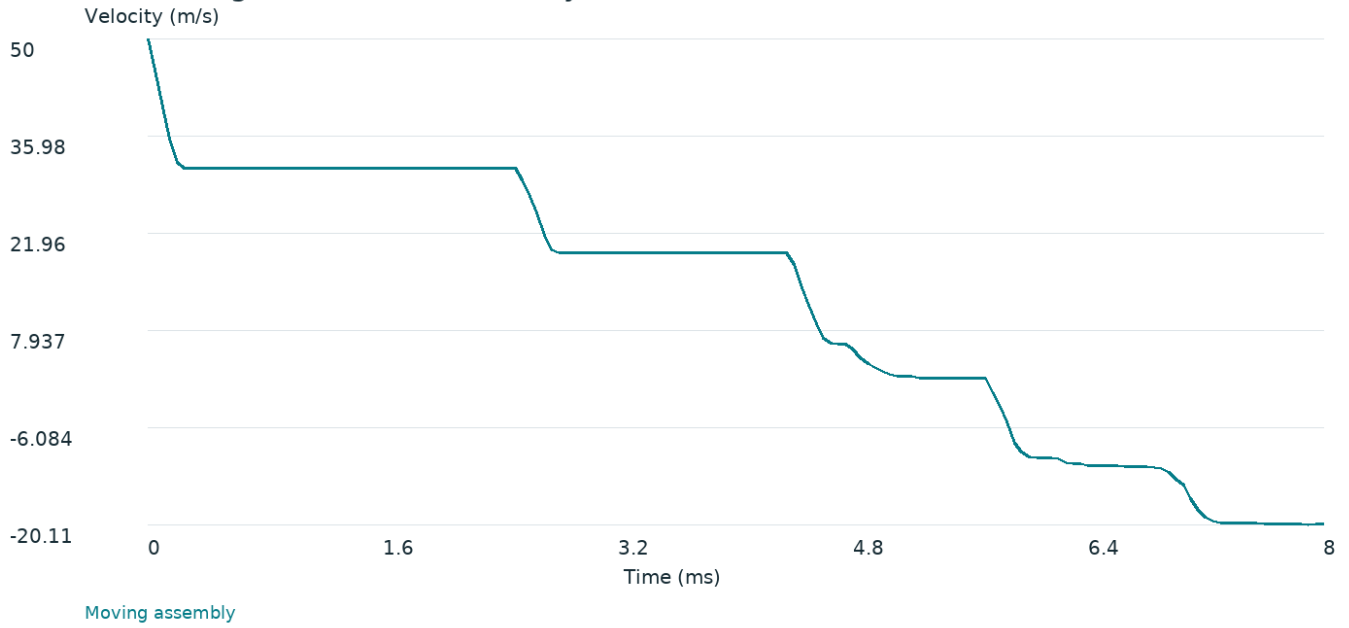
Recorded run status	completed_unassessed	
Recorded reached simulation time	7.99999	ms
Recorded requested simulation time	8	ms
Recorded solver wall time	4501.844	s
Recorded termination state	normal_termination_confirmed	

Separate solve charts



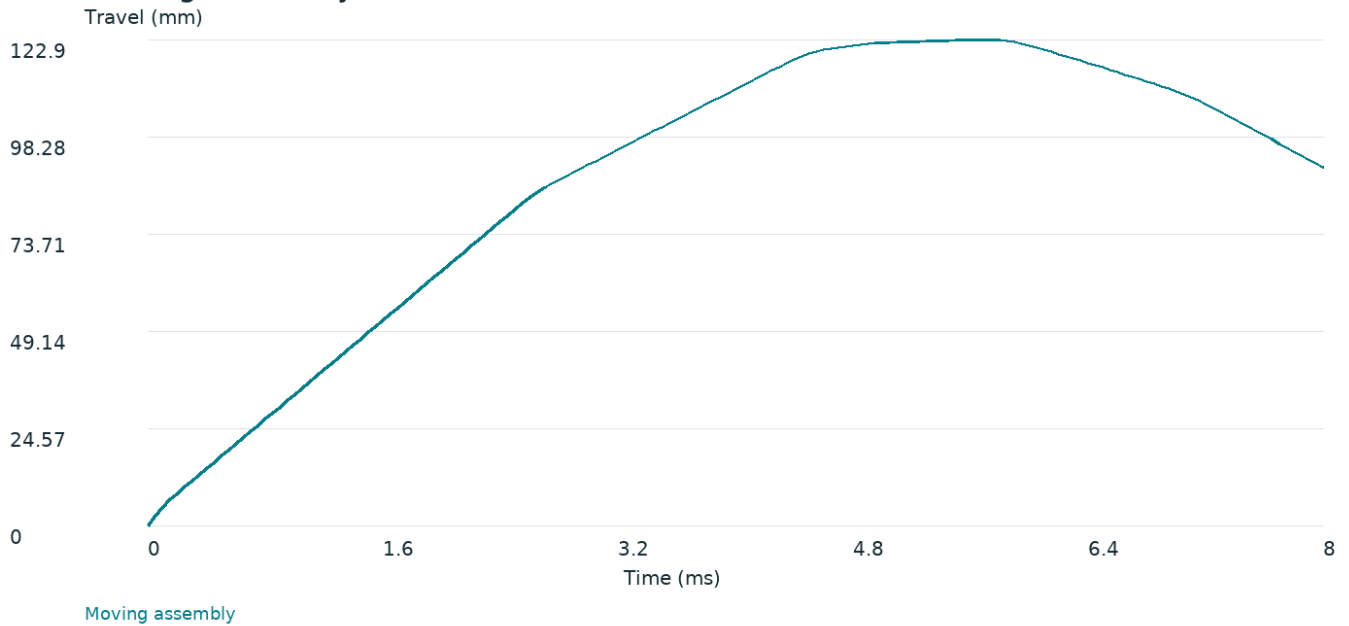
Contact force resultant. projects/local_closures_impact_20260927_01/cases/stem-5p5-5mm-v1/assessment/contact_histories.csv

Mass-weighted forward velocity



Mass-weighted forward velocity.
projects/local_closures_impact_20260927_01/cases/stem-5p5-5mm-v1/assessment/time_histories.csv

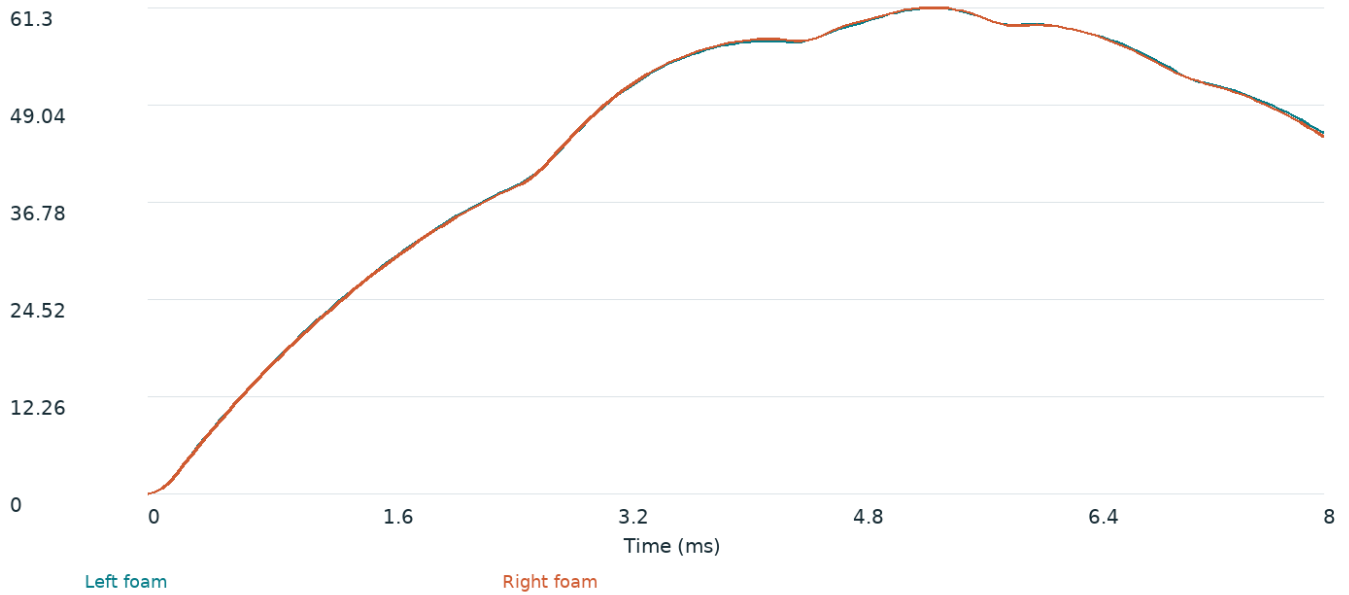
Moving assembly travel



Moving assembly travel. projects/local_closures_impact_20260927_01/cases/stem-5p5-5mm-v1/assessment/time_histories.csv

Foam compression

Compression (%)



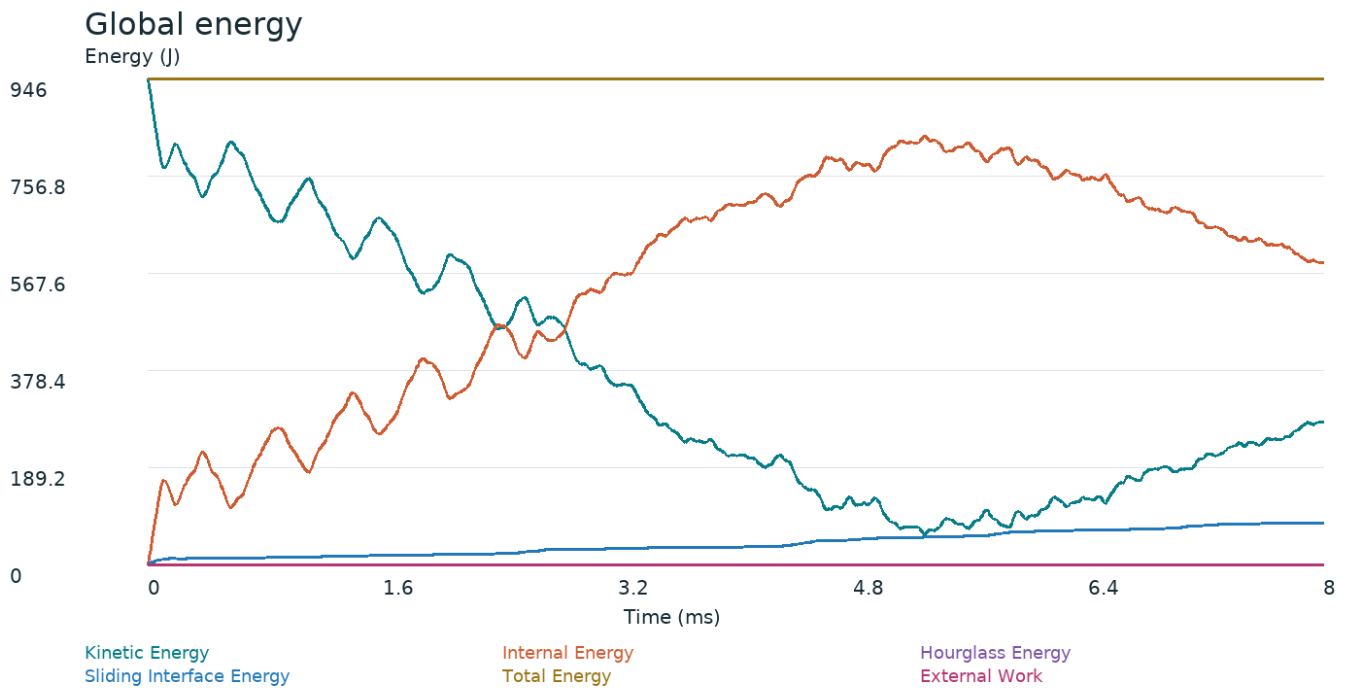
Foam compression. projects/local_closures_impact_20260927_01/cases/stem-5p5-5mm-v1/assessment/time_histories.csv

Regional P99 von Mises stress

Stress (MPa)



Regional P99 von Mises stress.
projects/local_closures_impact_20260927_01/cases/stem-5p5-5mm-v1/assessment/stress_histories.csv



Global energy. projects/local_closures_impact_20260927_01/cases/stem-5p5-5mm-v1/assessment/global_energy.csv

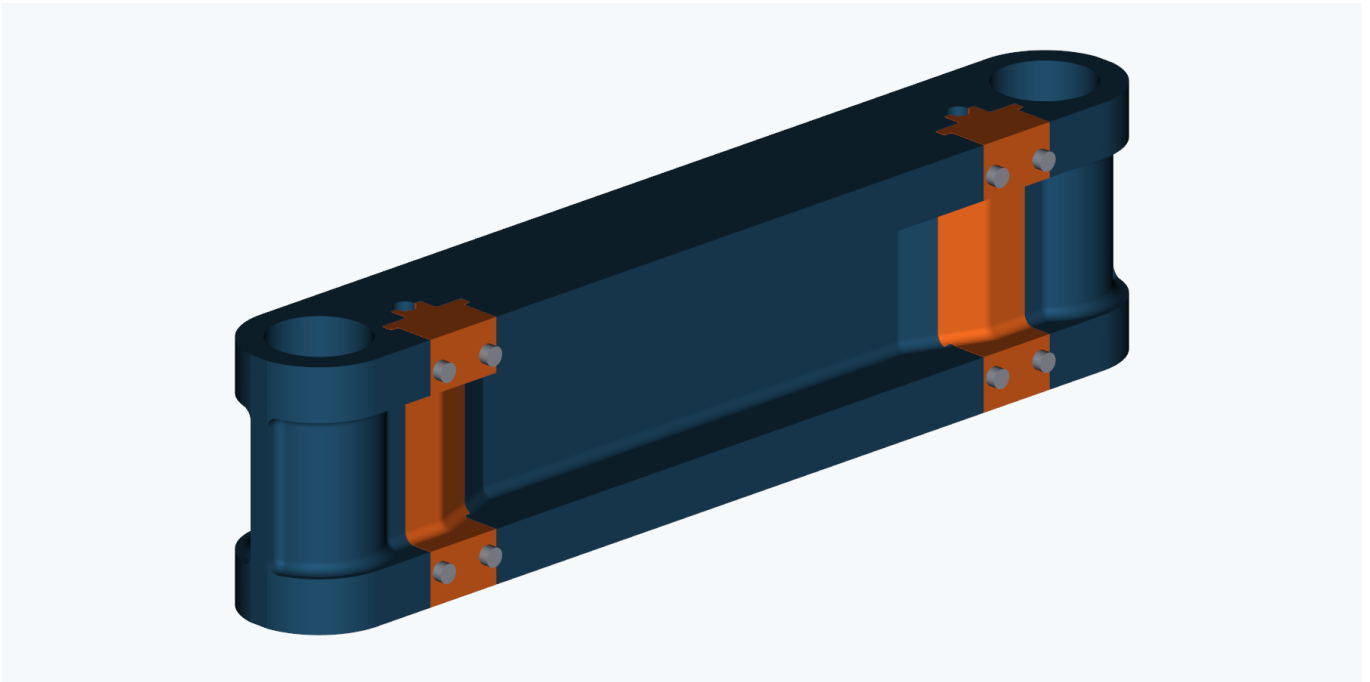
Results

Local cable-closure pusher impact: bounded geometry iteration

Geometry: original continuous body with two local T closures; revised 5.5 mm closure stem. **Solver:** ANSYS LS-DYNA explicit, 50 m/s, 8 ms. **Numerical screening:** all listed single-case checks passed; mesh and timestep convergence remain unresolved. **Engineering status:** provisional elastic demand, not physical survival qualification.

The supplied parent geometry stopped at **0.334 ms** when an element in the right closure's thin front cable wall drove the timestep to 1.76×10^{-15} s. The revised geometry completed the requested event. The wider stem therefore improved numerical robustness, but the matched early stress screen shows **higher** closure P99 demand. A second, narrower plug trial also stopped early at 0.178 ms. **Neither geometry revision is accepted as a stress improvement; the supplied parent CAD remains the next optimization seed.** The original parent has no valid full-event result to compare against this full run.

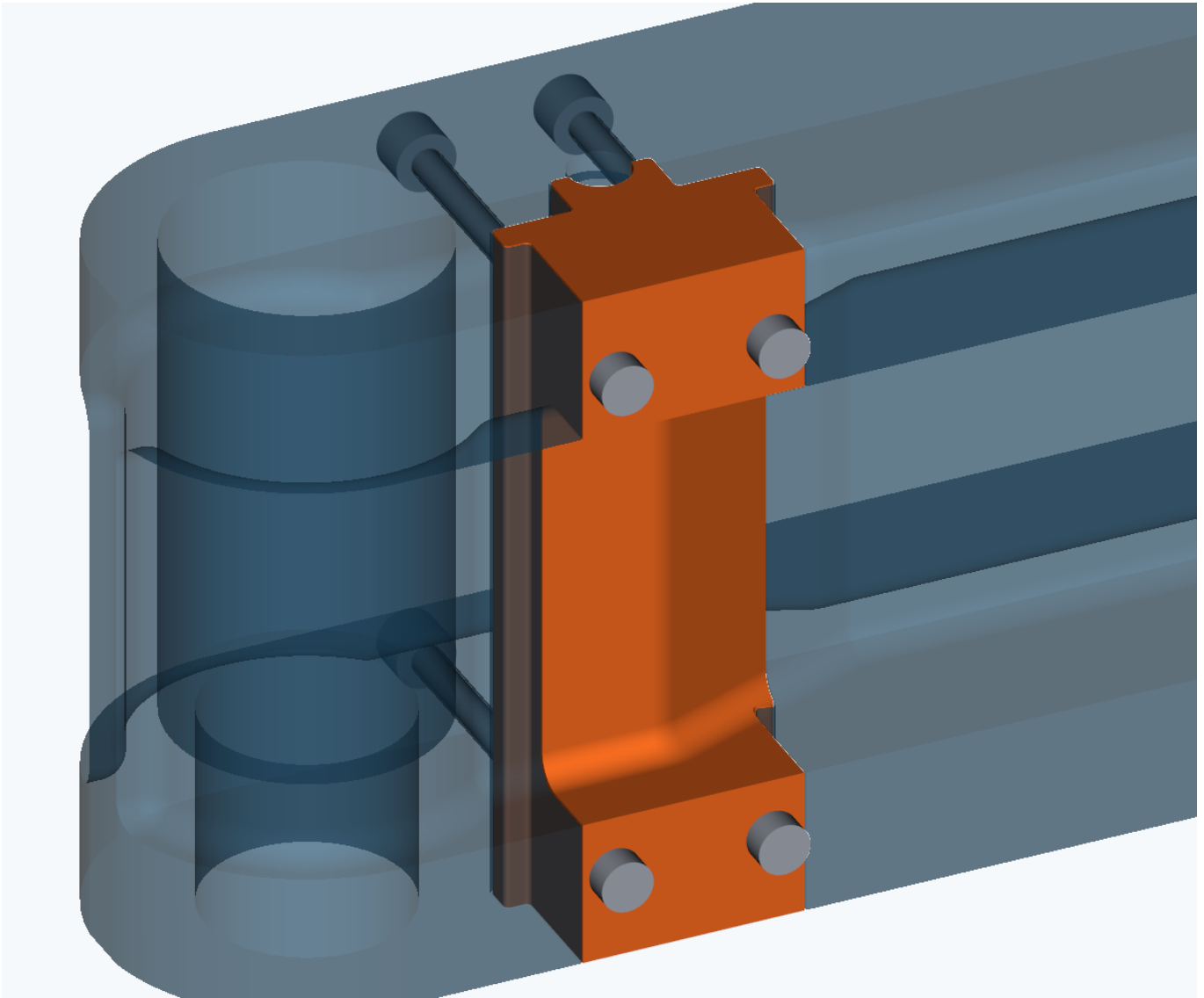
Geometry and load path



Revised assembly

The revised assembly keeps one printed main body, separate left/right closures and eight generic bolt/nut envelopes. The two motor bores remain at $X=\pm 133.25$ mm; cable guides remain at $X=\pm 101.6$ mm with 9.8298/6.75 mm stepped openings. The closure stem half-width rises from 4.3 to **5.5 mm**, with a matching body slot half-width of 5.7 mm. T shoulders, bolt axes and exterior fit remain fixed. The printed CAD mass is **561.182 g**, down 0.103 g from the supplied parent; hardware envelope mass is 34.002 g. All 19 STEP solids reimport and all three parts pass actual Bambu X1C slicing and generated extrusion-path checks. These checks do not establish printed strength or real hardware capacity.

The delivery contains the wider-stem STEP assembly, the narrow-plug trial assembly and the retained parent STEP assembly. Their separate component STEP files and build parameters are packaged alongside them. The parent is selected for the next iteration because neither revision demonstrated lower closure demand through a completed comparable event.



T closure detail

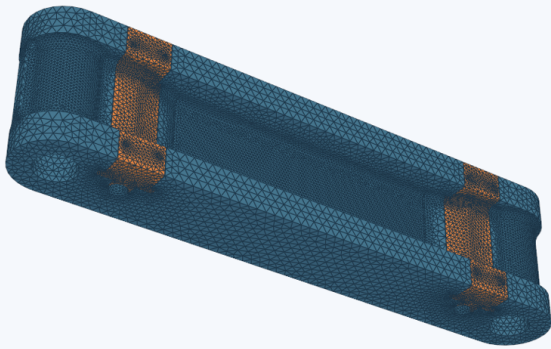
The model uses three independent flexible solids, frictional body/closure contact, and eight ideal clamped bolt collars carrying the CAD-derived generic hardware mass. Two 80 g motor masses are distributed over the bore nodes. The printed material is the prior study's isotropic elastic proxy (1060 kg/m³, E=1806.6 MPa, ν =0.35). The carrier and motor housings are excluded. The cable guides use eight transverse weighted-average equations and leave axial motion free; real cable compliance, diameter, tension and slip are not modeled.

The provisional fixture deliberately repeats the latest completed clamshell simulation: two initially stationary 41.275 × 152.4 × 8 mm aluminum spreaders at X=±152.3825 mm, 203.2 mm unconfined foam, 0.1 mm initial contact gap, friction 0.1 at pusher/spreader and closure interfaces, and fixed axial rear foam faces. The synthetic MAT057 foam has 54.46 kg/m³ density and 0.30337 MPa at 25% compression; it is **not** the proposed 30 kg/m³ foam product or a calibrated crush curve. The 346.04 mm spreader outside span also differs from the new nominal 350 mm stop span. This fixture is for direct comparison with the prior numerical model.

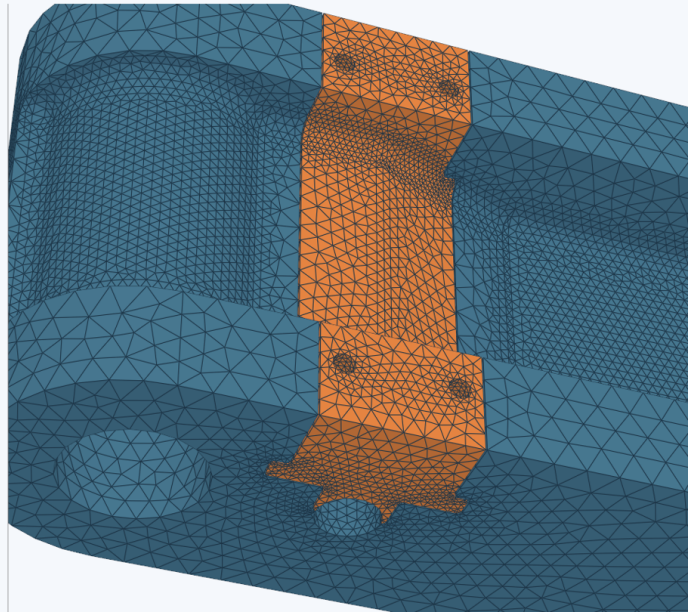
Solved mesh

Current solved tetrahedral mesh

321,831 tetrahedra | mean-ratio quality: min 0.074, P1 0.579, median 0.864 | 0 inverted



Right cable closure detail



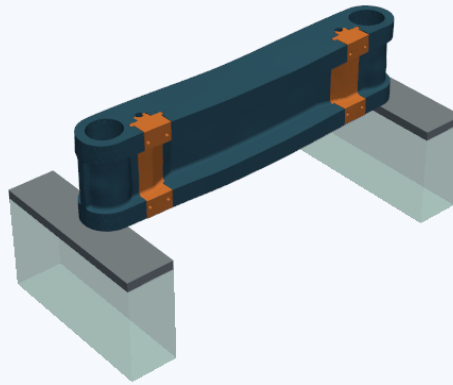
Actual tetrahedral surface mesh of the three printed parts

The completed solve used **321,831 first-order tetrahedra**: 188,650 in the body and 66,839/66,342 in the left/right closures. Body and closure target sizes were 5 and 3 mm. The tetrahedral **mean-ratio quality** (0 = degenerate, 1 = regular) had minimum **0.074**, first percentile **0.579**, and median **0.864**; 8 elements were below 0.1, with 0 inverted elements. The image shows the surface faces of the actual volume mesh. See the quality calculation and mesh hashes and mesh summary. This quality screen does not establish mesh convergence.

Completed event

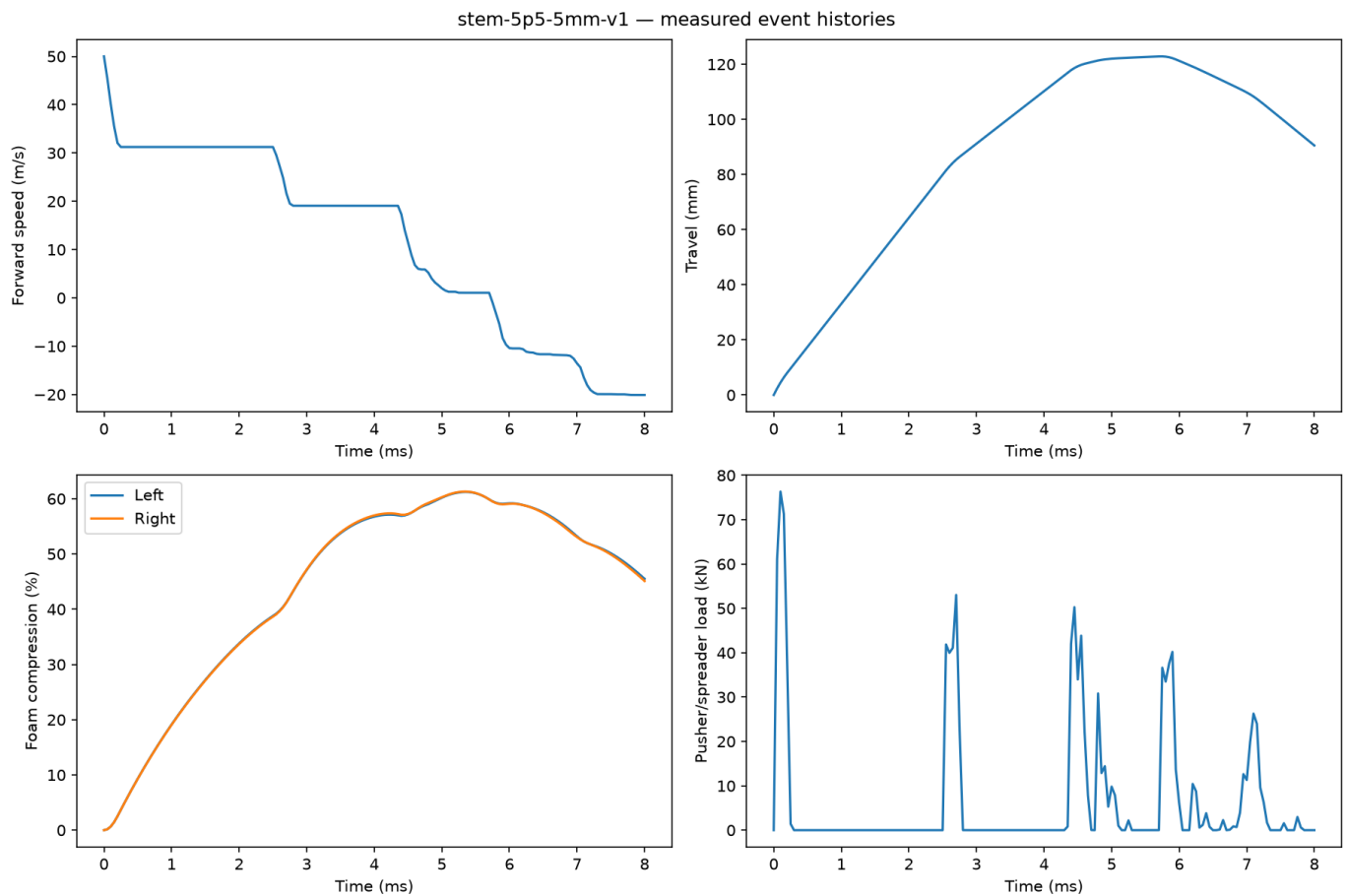
Speed -3.14 m/s
Compression L/R 59.3% / 59.2%
Travel 122.8 mm
Load 33.5 kN
Time 5.800 ms

- ▲ Assumed foam
- ▲ Aluminum spreader
- ▲ Body
- ▲ Left closure
- ▲ Right closure

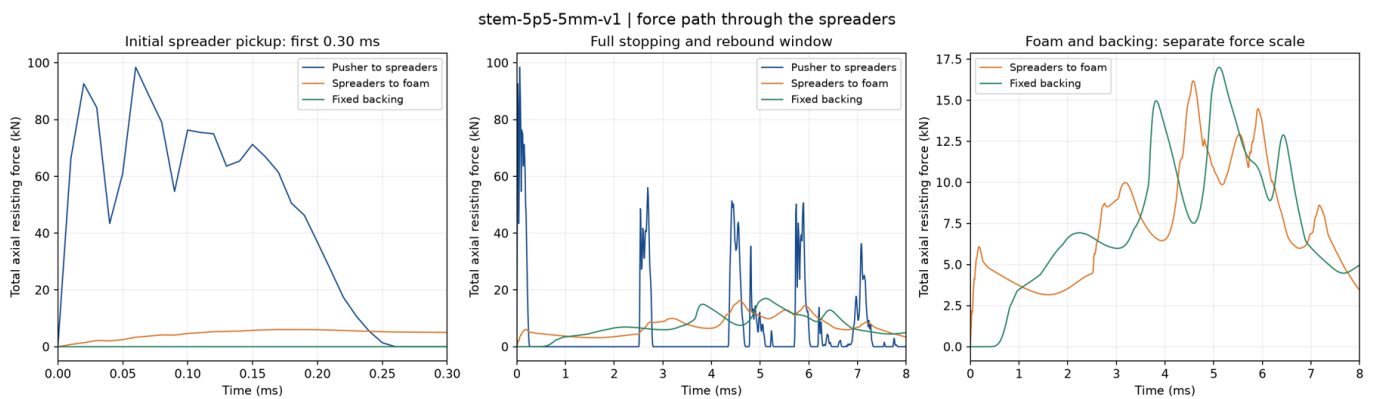


Pusher and fixture shortly after first stop, at 5.8 ms

The motion animation shows measured signed speed, spreader compression, travel, contact load and simulation time. Contact-force peaks can be briefer than the GIF frame spacing. **First mass-weighted stop:** 5.726 ms at 122.84 mm. **Maximum moving travel:** 122.85 mm. **Peak left/right foam compression:** 61.23% / 61.30%. **Final forward speed:** -20.11 m/s (negative means rebound). **Peak summed pusher/spreader contact load:** 98.45 kN at 0.060 ms. Maximum nonrigid displacement was 7.08 mm.



Time histories



Native contact and backing force path

The initially resting aluminum spreaders carry their own inertia. The native histories separate the initial **98.45 kN pusher pickup spike** from the peak **16.18 kN spreader/foam** contact at 4.580 ms and **17.01 kN backing** reaction at 5.110 ms. These are fixture-model force histories, not support-frame strength ratings. See native force peaks and backing history.

The exact histories, fixture settings, source and deck hashes are in time histories, contact histories, model, setup audit and execution record. Raw solver databases remain in the study case directory.

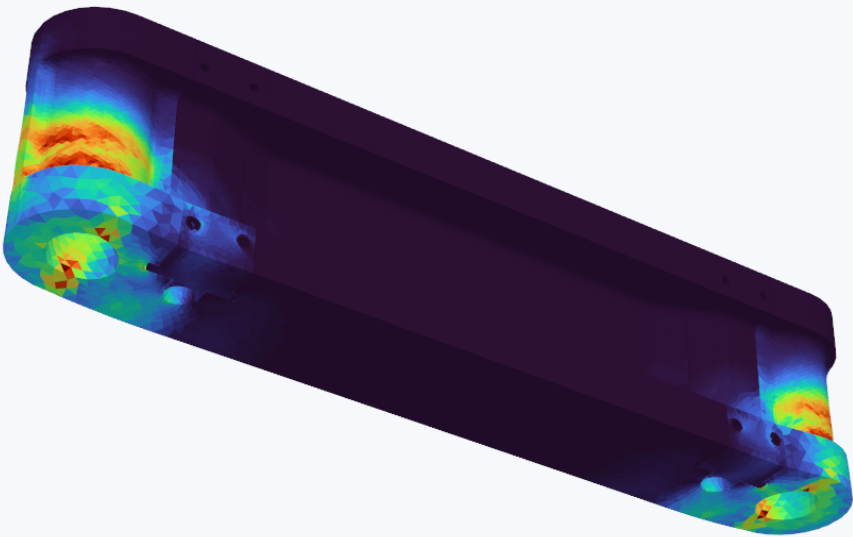
Stress and geometry implications

The values below are the maximum across saved states of **initial-volume-weighted element P95/P99 von Mises stress** in each region. They are elastic demand measures; isolated maxima and end-lip elements are strongly mesh dependent. Stress frames use a fixed 0–160 MPa saturated turbo display, and the camera follows the measured pusher translation to keep the whole part in view. Raw values remain in the stress histories.

Region	Peak P95 (MPa)	Peak P99 (MPa)
impact pod roots	129.33	161.41
motor seats	138.46	176.82
T slot body	79.66	122.41
closure left	57.64	169.54
closure right	50.44	175.73
bolt ligaments	47.76	83.52
central bridge	76.62	84.89
body	85.28	128.68

The highest raw saved element is part 3 at (96.6, 0.6, 0.1) mm, 1374.4 MPa. This isolated value is not a material-strength prediction. The model has no plasticity, fracture, bolt slip or printed anisotropy; very high local elastic stress signals that this idealization may have left its useful range.

stem-5p5-5mm-v1 0.050 ms



0-160 MPa display saturation; raw frame peak 197.8 MPa elastic proxy

Early impact stress

stem-5p5-5mm-v1 0.050 ms

TOP (+Z)



0-160 MPa display saturation; raw frame peak 197.8 MPa elastic proxy

Direct top view

stem-5p5-5mm-v1 0.050 ms

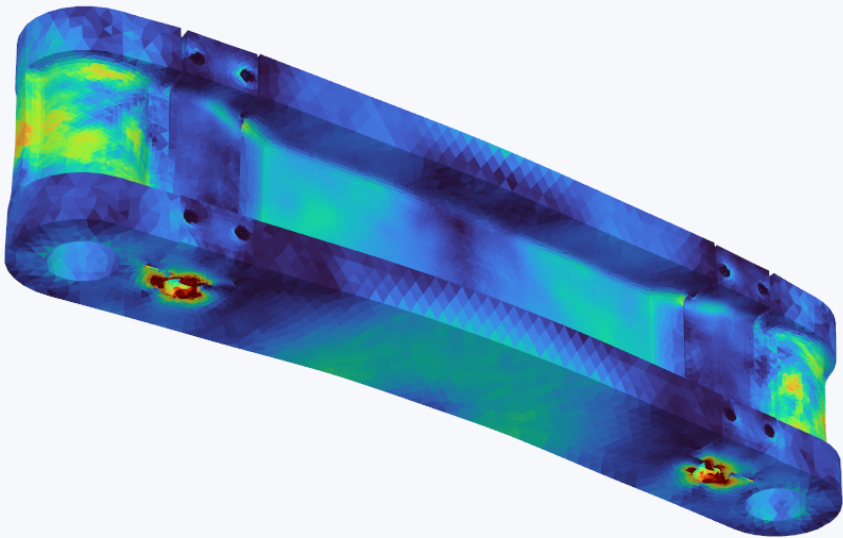
IMPACT FACE (-Z)



0-160 MPa display saturation; raw frame peak 197.8 MPa elastic proxy

Direct impact-face view

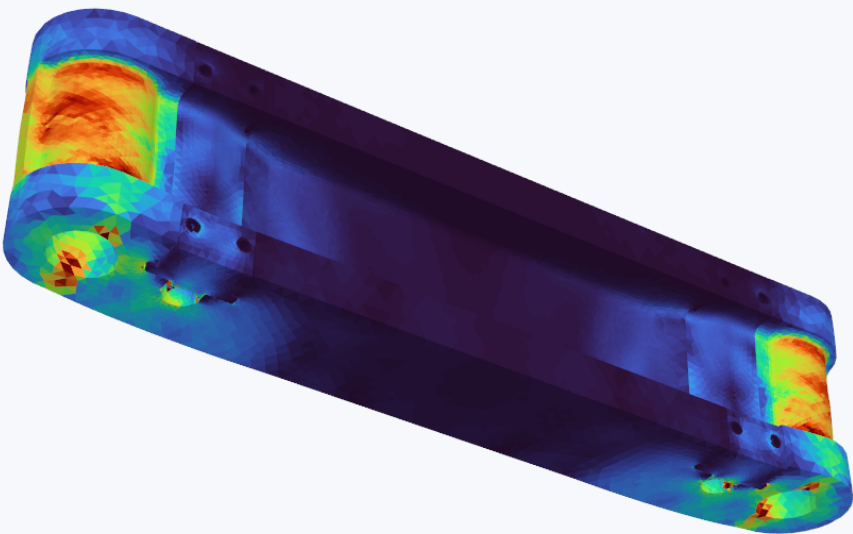
stem-5p5-5mm-v1 0.850 ms



0-160 MPa display saturation; raw frame peak 1254.5 MPa elastic proxy

Closure stress

stem-5p5-5mm-v1 0.100 ms



0-160 MPa display saturation; raw frame peak 271.0 MPa elastic proxy

Later body stress

stem-5p5-5mm-v1 0.100 ms

TOP (+Z)



0-160 MPa display saturation; raw frame peak 271.0 MPa elastic proxy

Direct later top view



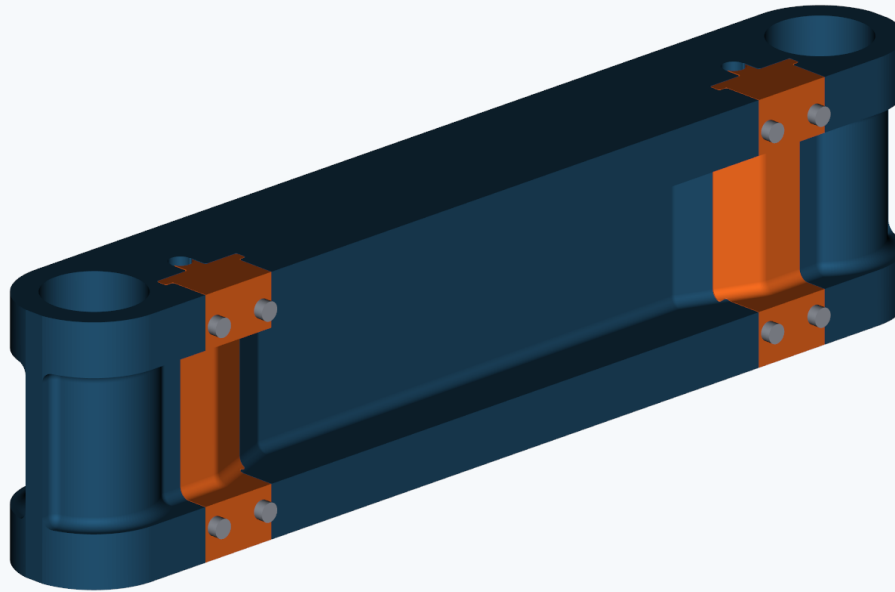
0-160 MPa display saturation; raw frame peak 271.0 MPa elastic proxy

Direct later impact-face view

The **closure cable-opening lips** remain the strongest local concern. The **motor pod impact roots and motor seats** carry broad early regional stress. The central bridge has lower aligned-impact stress demand, but asymmetric impact and torsion are not assessed, so cutting it back now would be premature. The T roots and bolt ligaments require local refinement and real fastener bearing/preload checks.

Comparison and qualification

The parent and revision share only a valid **0-0.299 ms** window. In that window, closure-left P99 rises +54.4% and closure-right P99 rises +27.9%; impact-pod-root P99 changes +0.3%. These are directional coarse-mesh observations, not a qualified stress ranking. The parent then collapses numerically. See the early-window comparison.



Narrower plug trial

The independent **narrow-plug** trial reduced the closure stem half-width to 3.2 mm and body slot half-width to 3.6 mm while preserving the cable installation sweep. It passed STEP and X1C toolpath checks, but a right closure element near the rear guide end (initial $Z \approx 73.5$ mm) drove the timestep to 2.66×10^{-15} s at 0.178 ms. Its execution record, partial metrics and element diagnosis are diagnostic evidence only. It never reached the stopping event and cannot be ranked against the completed run.

Single-case numerical check	Result
normal termination	PASS
requested time reached	PASS
binary matches nodout	PASS
moving mass consistent	PASS
initial energy error below 0p5pct	PASS
energy balance below 5pct	PASS
hourglass below 5pct	PASS
nonfriction contact energy below 5pct	PASS
contact impulse error below 5pct	PASS
no axial spc on moving or guide	PASS
no detected deletion	PASS
positive foam jacobians	PASS

Single-case numerical check	Result
first stop captured	PASS
foam compression below 80pct	PASS
added mass below 0p1pct nominal	PASS
guide equations satisfied	PASS

Energy balance error is 0.015% of nominal initial kinetic energy; peak summed absolute nonfriction contact energy is 0.386%; added mass reaches 0.038 g. These screens assess the run's numerical behavior only. The new closure has no mesh/timestep sensitivity result and no physical foam/material/fastener correlation, so design ranking and survival remain unqualified.

Next geometry decision

1. **Redesign the removable guide as a thicker load-bearing insert or split sleeve** around the cable, with a continuous front/rear lip and explicit bearing path into the main body. Retain the cable opening and installation path. The narrow printed end lips should not carry the idealized guide reaction through one thin edge.
2. **Rework fastener load transfer** with selected bolts, washers or inserts, adequate edge distances, and modeled shank bearing/preload. Widening the stem alone consumes material beside the current hole pattern; keep the present wider-stem variant as an unsuccessful experiment.
3. **Round/gusset the motor pod-to-bridge transition** and verify spreader contact reach under the actual 350 mm fixture. This is the next credible region for stress reduction after the guide is stabilized.
4. **Consider mass removal from the central bridge only after left/right-leading and roll cases.** The aligned run shows lower central demand but does not establish torsional reserve.

The next agent should use CONTINUE.md, select real foam and aluminum hardware when available, model compliant cable contact and hardware bearing, and rerun matched parent/revision cases before accepting a design. Per the user's updated standard, that next printed-part analysis mesh is to be generated in **Altair HyperMesh** with conforming **hexahedral volume elements and quadrilateral surface faces**. The current tetrahedral result remains historical evidence and cannot be relabeled as a hex solve.

Authored source: projects/local_closures_impact_20260927_01/report/REPORT.md. Original narrative and values retained. Animation figures and their captions are excluded from this PDF; animations remain available in the standalone HTML.